

Example 6: NR Classroom Laboratory, VAV-DOAS with Four-Pipe Fan Coils and Air-to-Water Heat Pump

SYSTEM TAG: T24STD-DOAS-FPFC-AWHP-HRV

PROJECT VISION

California Title 24 compliance modeling requires translating mechanical system designs into CBECC-Com 2025 software objects, a process for which little practical guidance exists. Developed as part of the *Addressing Complex HVAC Systems in Code Compliance* project, funded by Southern California Edison through the CalBEM working group, each example walks through how a recognizable HVAC system type translates into CBECC-Com objects, what Title 24 2025 Part 6 requires, and what assumptions were made.

PREPARED FOR

Southern California Edison
CalBEM Working Group 3

PREPARED BY

A2 Efficiency & West Monroe

Example 6: NR Classroom Laboratory, VAV-DOAS with Four-Pipe Fan Coils and Air-to-Water Heat Pump

System Tag: T24STD-DOAS-FPFC-AWHP-HRV

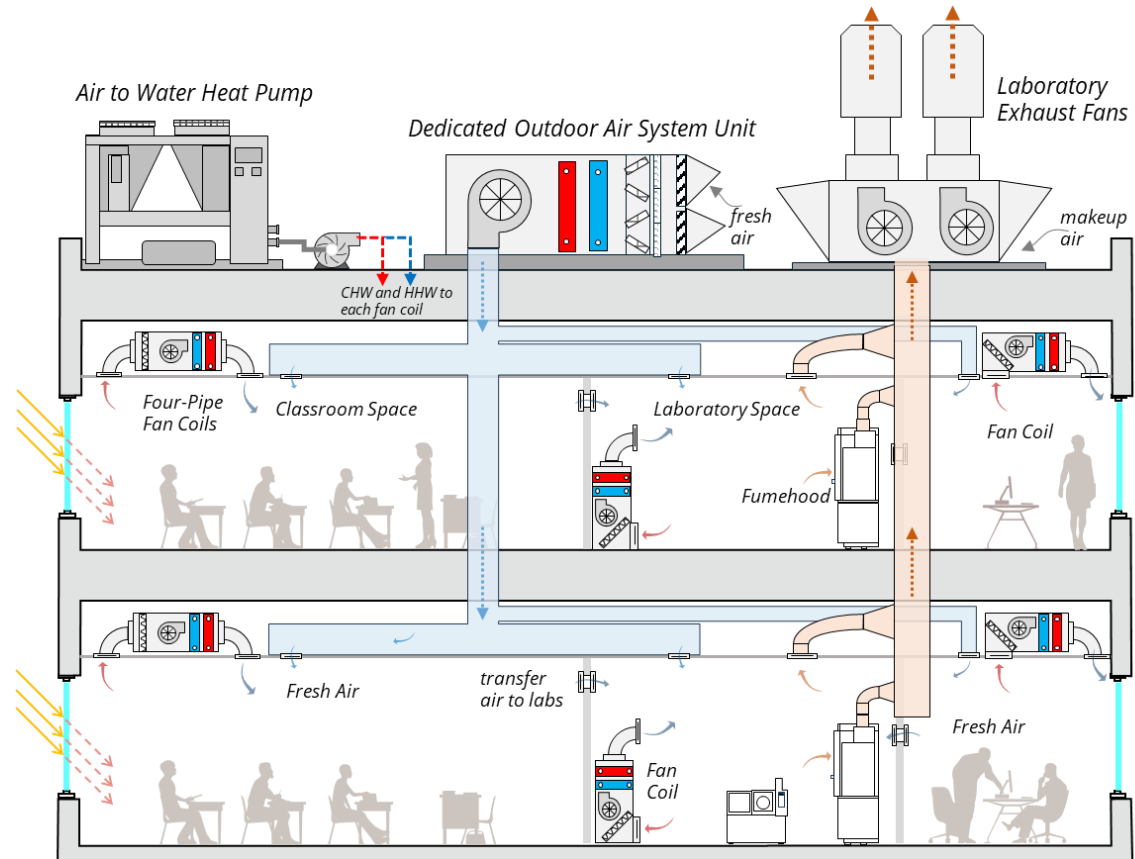
Title 24 2025 STD:140.4-Aii

Mechanical Description

The building is conditioned by a dedicated outdoor air system (DOAS) with distributed four-pipe-fan coil units (FPFC), providing heating and cooling. A central plant, consisting of an air-to-water heat pump, produces hot and cold water conditioning the ventilation air and each thermal space.

Chilled water as well as hot water for heating and cooling are generated by an air-to-water heat pump. Fresh air is provided by a central DOAS unit, transferred from classrooms and offices to laboratory spaces, and exhausted through a central laboratory exhaust system. The laboratory spaces include fumehoods to demonstrate how exhaust driven ventilation systems are balanced.

Laboratory exhaust is handled by a dedicated variable speed exhaust fan system capable of wide turndown to track actual exhaust demand across varying fume hood and general exhaust loads.



The diagram shows a central DOAS unit supplying tempered ventilation air to each zone. A separate fan coil is provided in each thermal zone, with ventilation air supplied to each zone and transferred to the laboratories for cascaded airflow.

Mechanical Details

The system was applied to the custom lab classroom prototype, with 25 thermal zones.

Ventilation: 100% outdoor air delivered to each zone via DOAS. Air is cascaded through the classrooms to the laboratories and exhausted with additional exhaust at the DOAS.

Ventilation control: Variable volume DOAS with zone-level VAV terminals in rooms where DCV is required and CV terminals in others.

Cooling: Four-pipe fan coil units for space cooling with chilled water from air-cooled screw chiller; DOAS chilled water coil for ventilation air pre-cooling.

Heating: Four-pipe fan coil units for space heating with hot water from air-to-water heat pump; DOAS hot water coil for ventilation air tempering.

Capacity control: Variable chilled and hot water flow at fan coil units with variable-speed fans; DOAS operates at variable ventilation airflow tracking zone demand

Economizing: No economizing

Heat Recovery: Sensible and latent heat recovery through energy recovery ventilator serving the DOAS.

Mechanical Capacities and Sizing

The following tables show the capacities for major components in this example model for the DOAS ventilation unit and central plant heating and cooling devices.

The ventilation system was sized based to meet total outdoor air requirements across all zones, with heating and cooling coil capacities matched to peak ventilation loads.

A exhaust fan was included in the lab exhaust air system only. To enable the DOAS air system to function only a supply segment and fan were configured without any exhaust or return air segment.

Air Systems	Sys Type	Fan Ctrl	Supply CFM	Min CFM (VSD)	Supply Density (CFM/sf)
DOAS-AHU Supply	DOAS	VSD	15,879	10,177	0.59
DOAS-AHU Exhaust	DOAS	VSD	5,294	5,294	0.30
Lab Exhaust Fan	Exhaust	VSD	10,585	4,883	0.39

Equipment	Plant Type	Rated Capacity (Btu/h)	Capacity / Building Area (Btu/sf)	EER	IPLV	COP
Air Cooled Chiller	ChilledWater	989,261	36.6	15	21	n/a
Air-to-Water HP	HotWater	308,705	11.4	n/a	n/a	2.5

Four-pipe fan coil units were sized with each zone's heating and cooling loads, with independent chilled water and hot water coil capacities assigned separately to reflect the decoupled nature of the four-pipe configuration.

Central plant equipment: the air-cooled screw chiller and air-to-water heat pump were sized to the peak loads of the FCUs and DOAS coils. Chiller and heat pump efficiencies were specified based on capacity and minimum allowed efficiency requirements per Title 24 2025. DOAS heating and cooling coil components were assigned efficiencies consistent with code minimums, and heat recovery effectiveness was assumed to reflect typical energy recovery ventilator performance.

Code Compliance Software Configuration

DOAS System

The DOASVAV type delivers 100% outside air and includes:

- A chilled water cooling coil and hot water heating coil served by the central hydronic plant.
- A variable-speed drive centrifugal supply fan to modulate flow.
- An exhaust fan to balance flow.
- An outside air control for demand-based ventilation.
- A HeatRecovery object to recover energy from exhaust.

VAV Terminal Units

VAVNoReheatBox terminal units distribute ventilation air to zones. Zones with CO₂-based control requirements have the lower limit configured to enable modular flow.

Central Hydronic Plant

The chilled water plant is modeled as an air-cooled screw chiller on a primary variable-flow loop. The hot water plant uses an air-to-water heat pump. Both loops serve DOAS coils and fan coil units.

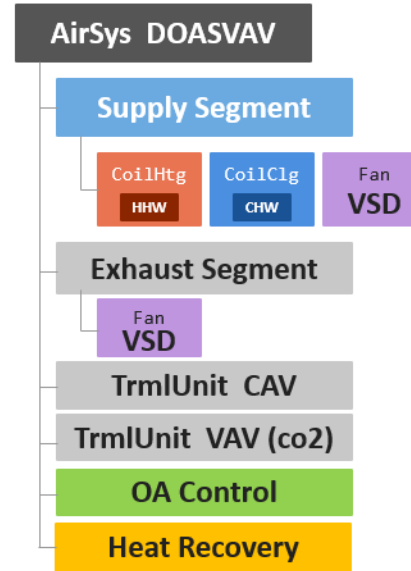
Lab Exhaust System

The laboratory exhaust system uses a variable-speed exhaust fan sized to peak

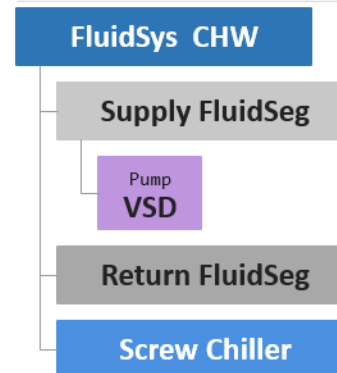
exhaust demand. Supplemental zone fans serve exhaust zones such as restrooms.

Building Systems

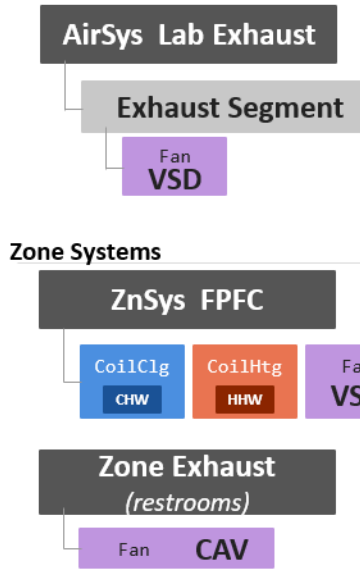
Air System, One Unit



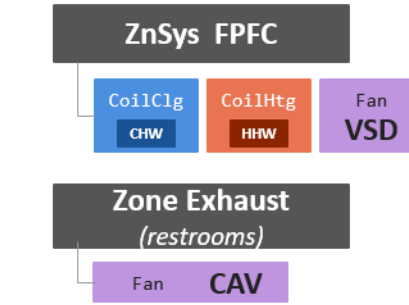
Plant Systems



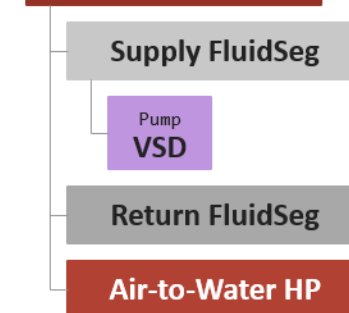
Air System, One Unit



Zone Systems



FluidSys HHW



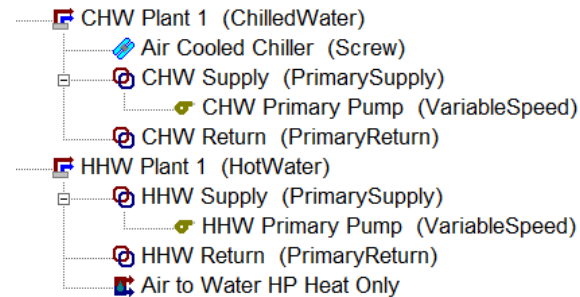
The system configuration is shown from CBECC 2025.2 for the DOAS -VAV-FPFC (CHW/HHW) system components.

The plant systems are depicted for the chilled water and hot water systems.

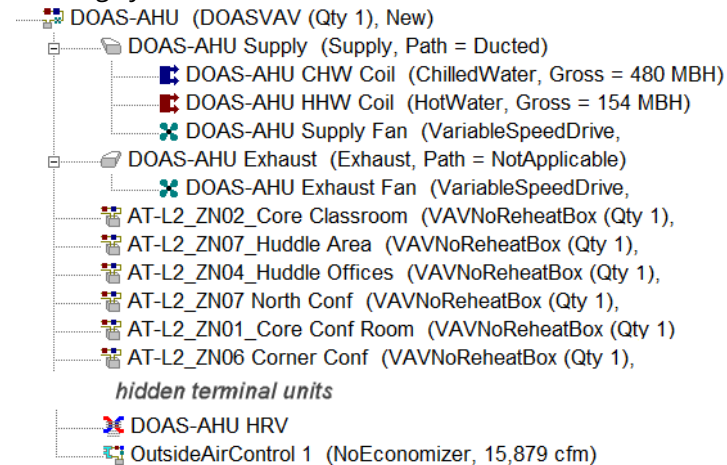
The DOAS air system is shown with each coil, fan, and terminal units, and finally the heat recovery device and outdoor air controller

The laboratory exhaust system, an air system, is shown with the single exhaust fan type. The zone system exhaust fans and a sample zone fan coil system are also shown with their individual coils and fans.

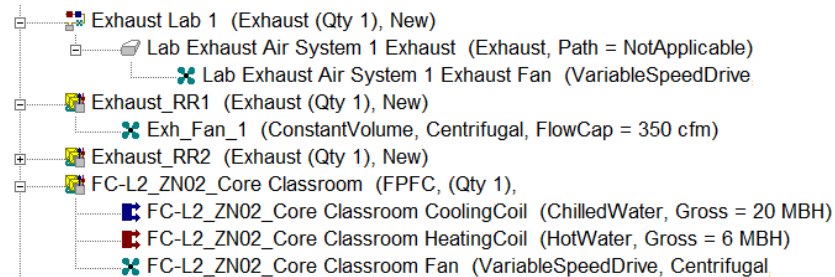
Plant Systems



Building Systems



Continued



CBECC Object Locations

Key screenshots of the system for the ventilation unit, central plant components, and the laboratory exhaust are shown below.

Plant System

The heating plant consists of an air-to-water heat pump, supplying hot water at the fan coils and DOAS unit HW coil.

Coil Type	HeatPump (air-to-water)
Capacity	308,705 Btu/hr (25.7 tons)
Capacity / Area	11.42 Btu/sf
COP	2.5 (heating)
Flow Rate	61.72 gpm
HP Lockout	36°F

The cooling plant consists of an air cooled chiller serving the same fan coils and a DOAS CHW coil.

Coil Type	Air Cooled Chiller (Screw)
Net Capacity	989,261 Btu/hr (82.5 tons)
Capacity / Area	36.56 Btu/sf
EER	15
IPLV	21
Flow Rate	164.8 gpm
Entering and Leaving Temp	56°F/44°F

Currently Active Heat Pump: **Air to Water HP Heat Only**

Name: **Air to Water HP Heat Only** Status: **New**

Rated Ent/Lvg Wtr Temp: **120 F/140 F**

Rated Heating Capacity: **308,705** Btu/h

Flow Rate: **61.72** gpm

Rated Heating COP: **2.50**

Min Part Ld Ratio: **0.25**

Cprsr Lockout Temp: **36.00**

Compressor location: **Outdoor**

Load Inlet FluidSeg: **HHW Return**

Load Outlet FluidSeg: **HHW Supply**

Defrost Control: **Timed**

Defrost Temp.: **36.0** °F

Supplemental Boiler: **- none -**

Currently Active Chiller: **Air Cooled Chiller**

Name: **Air Cooled Chiller** Status: **New**

Type: **Screw**

Condenser Type: **Air**

Input Fuel: **Electric**

Rtd. Capacity: **989,261** Btu/h

Flow Rate: **164.8** gpm

EntTemp -Dsgn: **56.0** °F

LvgTemp -Dsgn: **44.0** °F

EER: **15.0** - IPLV: **21.0**

Min Unload Rat: **0.150** frac

Min Part Ld Rat: **0.150** frac

Evap. Inlet FluidSeg: **CHW Return**

Evap. Outlet FluidSeg: **CHW Supply**

☒ Evaporator Has Bypass?

DOAS Air System

The DOAS is a 100% outdoor air system configured as a DOASVAV type, enabling ventilation airflow to modulate at the zone depending on the terminal unit types.

The air temperature is maintained at neutral conditions, which has design supply temperatures of 55 F and 60 F for cooling and heating.

To maintain temperature at night, the local zone fan coils are used, keeping the ventilation system off.

Air System Terminal Units

Air terminal units are created under the DOAS Air System for each thermal zone ventilation delivery. The terminal units are configured for ventilation only, providing peak flow for fully occupied and a lower level for minimum ventilation allowed based on CO2 sensors.

The screenshot shows the 'Air System Data' window for 'DOAS-AHU'. Key fields are highlighted with red boxes: 'Name: DOAS-AHU', 'Type: DOASVAV', 'Status: New', 'Sub Type: Packaged3Phase', 'Count: 1', 'Fan Control: Continuous', 'Night Cycling: CycleZoneFansOnly', 'Design Supply Air Temp: 55.0 °F (Cooling) and 60.0 °F (Heating)', 'Net Capacity*: 480,181 Btu/h (Cooling) and 154,344 Btu/h (Heating)', 'Fan Position: DrawThrough', 'Supply Temp Control: OutsideAirReset', 'Fixed Supply Temp: 55.0 °F (Cooling) and 60.0 °F (Heating)', 'Design OA Flow: 15,879 cfm', 'Control Type: DDCToZone', 'Reheat Control: SingleMaximum', 'Supply Flow: 15,879 cfm', '100% OA System' (checked), 'Optimum Start' (checked), and 'Complex Mechanical System?' (checked). The 'Sizing Parameters' section shows 'Design Flow/Area: 1.000 cfm/ft2' and 'Design Flow/Ton: 400.0 cfm/ton'.

The screenshot shows the 'Terminal Unit Data' window for 'AT-L2_ZN02_Core Classroom'. Key fields are highlighted with red boxes: 'Name: AT-L2_ZN02_Core Classroom', 'Type: VAVNoReheatBox', '# of Terminal Units: 1', 'Status: New', 'Component Qty: 1', 'Zone Served: L2_ZN02_Core Classroom', 'Primary AirSeg Ref: DOAS-AHU Supply', 'Max. Primary Flow: 656 cfm, 0.52 cfm/ft2, 2.28 ACH', and 'Min. Primary Flow: 188 cfm, 0.15 cfm/ft2, 0.65 ACH'.

Thermal Zones

At each thermal zone, ventilation is configured from the DOAS unit and heating and cooling from the zones designated four pipe fan coil.

Nine zones have CO2 sensors based on occupant density and therefore also have minimum and maximum ventilation rates. All zones utilized a Variable Air Volume Terminal Unit for simplicity though only nine were configured with a lower airflow limit.

The screenshot shows the 'Thermal Zone Data' tab for the 'L2_ZN02_Core Classroom'. The 'Currently Active Thermal Zone' is 'L2_ZN02_Core Classroom' and the 'Schedule Group' is 'School'. The 'Name' is 'L2_ZN02_Core Classroom', 'Type' is 'Conditioned', 'Vent. Source' is 'Forced', and 'Description' is empty. The 'HVAC Zone Count' is 1, 'Floor Area' is 1,255.3 ft2, 'Zone Multiplier' is 1, and 'Occupants' is 43.4 people. The 'Supply Plenum Zone' and 'Return Plenum Zone' are both set to '- none -'. Under 'HVAC Systems', the 'Ventilation System' is 'DOAS-AHU' with priority 1, and 'Htg/Clg System 1)' is 'FC-L2_ZN02_Core Classroom' with priority 2. A checkbox 'Sys 1) Connected to DOAS' is present.

The screenshot shows the 'Ventilation and Exhaust' tab for the 'L2_ZN02_Core Classroom'. The 'Currently Active Zone' is 'L2_ZN02_Core Classroom'. Under 'Ventilation (Outdoor) Air', the 'Ventilation Standard' is 'T24-2025', 'Specification Method' is 'Maximum', and 'Control Method' is 'CO2Sensors' (highlighted with a red box). The 'System Has DDC To Zone' checkbox is checked. The 'Floor Area/Volume' is 1,255.3 ft2 / 17,260.9 ft3, and 'Occupancy' is 43.4 people / 34.6 people/1000ft2. The 'Exhaust' section is partially visible on the right.

Thermal Zone Exhaust Ventilation Control Air Terminal Units

For classrooms with laboratory exhaust, the second tab of inputs in the Thermal Zone Data object is a key source for definition. This tab defines the exhaust system serving this building as well as links the exhaust system to an exhaust heat recovery device, located in the supply side air system.

Zone System Four Pipe Fan Coils

The zone systems consist of four pipe fan coils, serving each thermal zone. Additional air terminal units supply the fresh air from DOAS. Even with systems where fresh air is routed to a fan coil, the proper configuration is to enable two elements serving each zone in CBECC, a terminal unit and a zone system.

Units were configured with cycling fans, which is the only modulating fan control allowed for these zone systems. Supply air temperature was left at default, supplying 55°F air in cooling, 95°F heating.

Fans are set to variable speed drive. Regardless of their assignment, in CBECC 2025 they are set to on off fans upon simulation and will cycle on and off as needed.

Thermal Zone Data | Ventilation and Exhaust | Temp Control and Sizing | Daylighting

Currently Active Zone: **L1_ZN03_South Open Office**

Ventilation (Outdoor) Air

Ventilation Standard: **T24-2025**

Specification Method: **Maximum**

Control Method: **Fixed** ☒ System Has DDC To Zone

Floor Area/Volume: **785.8** ft2 **10,280.6** ft3

Occupancy: **19.6** people **25.0** people/1000ft2

	Design	Code Min.
Total Ventilation (cfm):	1,021.5	1,021.5
(cfm/ft2)*:	1.300	1.300
(ACH)*:	5.962	5.962
Min. Ventilation (cfm):	685.4	4.000 ACH

*Flow rate shown is the total as a normalized value, not component values. See HELP for details.

Currently Active Zone System: **FC-L2_ZN02_Core Classroom**

Name: **FC-L2_ZN02_Core Classroom**

Type: **FPFC** Status: **New**

Sub Type: **NA** Count: **1**

Control Zone: **L2_ZN02_Core Classroom**

Description:

Fan Control:

Design OA Flow:

	Cooling	Heating	Supply Flow
Design Supply Air Temp:	55.0 °F	95.0 °F	
Net Capacity*:	19,837 Btu/h	6,376 Btu/h	1,88

*Reflects capacity of a single system if Count is >1. Heating capacity includes terminal units.

Four Pipe Fan Coil Heating and Cooling Coils

Each zone is served by a four-pipe fan coil unit with both a chilled water cooling coil and a hot water heating coil connected to their respective supply and return fluid segments. The unit operates with a variable speed drive centrifugal fan with a defined maximum and minimum flow capacity

Cooling Coil Data	Performance Curves
Currently Active Cooling Coil: FC-L2_ZN02_Core Classroom CoolingCoil	
Name:	FC-L2_ZN02_Core Classroom CoolingCoil
Type:	ChilledWater
Status:	New
Component Qty:	1
Inlet FluidSeg:	CHW Supply
Outlet FluidSeg:	CHW Return
Capacity (For single system/terminal if Component Qty > 1)	
Rated Gross Capacity:	19,837 Btu/h
Design Flow Rate:	3.3 gpm
Pressure Drop:	5.0 ft H2O

Heating Coil Data	Performance Curves
Currently Active Heating Coil: FC-L2_ZN02_Core Classroom HeatingCoil	
Name:	FC-L2_ZN02_Core Classroom HeatingCoil
Type:	HotWater
Status:	New
Component Qty:	1
Inlet FluidSeg:	HHW Supply
Outlet FluidSeg:	HHW Return
Capacity (For single system/terminal if Component Qty > 1)	
Rated Gross Capacity:	6,376 Btu/h
Design Flow Rate:	1.3 gpm
Pressure Drop:	5.0 ft H2O

System Efficiency Tables

Category	Proposed (FPFC/DOAS+CHW/HHW)	Standard (SZHP / PVAV / SZAC)
System Type	FPFC/DOAS central plant all zones (IT: SZAC)	SZHP: small zones PVAV: large open-office zones SZAC: IT
Cooling Efficiency	Central air-cooled chiller EER=15.0 / COP=4.40	SZHP: EER=11.4 / SEER=14 PVAV: EER=9.8
Heating Primary	Air-to-Water HP COP = 2.5 HHW to FPFC coils and DOAS	SZHP: air-source HP COP=3.2 / PVAV: gas HHW boilers 82% AFUE
Heating Supp.	None	SZHP: electric resistance at 35°F OAT PVAV: gas HHW reheat
Fan Type	VSD on ALL FPFC zones + VSD DOAS (0.35–2.29 W/cfm zone fans)	SZHP zones: CV (0.58–0.69 W/cfm) PVAV: VSD (1.10–1.11 W/cfm)
Economizer	No Economizer at zone level (DOAS pre-conditions OA; no free-cool cycle)	SZHP: FixedDryBulb 71°F on 11 zones PVAV: NoEconomizer
DCV	CO2Sensors on 11 zones Fixed on 16 zones	CO2Sensors on fewer zones Fixed on most
Supp Lockout	N/A, no supplemental heat, Defrost enabled at 36°F	SZHP: electric resistance adds at 35°F OAT HP lockout at 17°F

Relevant Energy Code Citations

Required Demand Control Ventilation

401.2.1.2.3 [Section 120.1(d)3]

Nine zones exceeded the occupancy density limit, requiring CO₂-based modulating ventilation airflow. Zone-level VAV terminals on the DOAS serve those zones.

Fan Power Budget - DOAS Unit

401.3.3.4 [Section 140.4(a)3E] and 401.3.17.1 [Section 140.4(p)]

The DOAS unit is required to comply with the prescriptive system requirements for a four-pipe fan coil system with DOAS. That section sets a maximum DOAS fan power of 0.77 W/cfm at design airflow. The modeled DOAS fan power is consistent with this prescriptive limit.

Fan Power Budget - Indoor Fan Coil Units

401.3.3.3 [Section 140.4(a)3D]

Indoor fan coil units were aligned with prescriptive code for the multi-zone heat pump system to a fan power of 0.35 W/cfm.

Air-Side Economizer Exemption - DOAS Unit

401.3.7 [Section 140.4(e)]

The DOAS unit operates at 100% outdoor air and therefore has no return air damper to modulate. An air-side economizer is not applicable to this system configuration.

Exhaust Air Heat Recovery

401.3.18 [Section 140.4(q)]

Outdoor air systems delivering 3,000 CFM or more at 70% or greater outdoor air fraction are required to include exhaust air heat recovery. The DOAS delivers 100% outdoor air above this airflow threshold, requiring the modeled heat recovery ventilator.

Minimum DOAS Cooling and Heating Efficiency

Section 140.4(a)3Aii, 140.4(a)3Aiii, or 140.4(a)3B

The DOAS cooling coil and heat pump heating coil are each required to meet minimum EER and COP thresholds for their capacity ranges. Both modeled coils exceed the applicable mandatory minimums.

Lab Hoods

The lab exhaust fans are required to meet thresholds per Table 120.1-C, Table 140.4-B, Section 140.9(c), Table 170.2-C. Lab exhaust fans are modelled according to the mandatory minimums.

Same Criteria as Example 1

- Duct Sealing and Leakage Testing
- Direct Digital Controls to Zone
- Optimum Start/Stop Controls

This example is one of eight developed as part of the *Addressing Complex HVAC Systems in Code Compliance* project. The full guide includes additional HVAC system types, alternative configurations, and working CBECC-Com model files. All resources are available through the CalBEM working group and EnergyCodeAce platform.

